

Einsendaufgaben – Aufgabe 5.4

Modul 61111: Mathematische Grundlagen

Aufgabe 5.4

a)

$$\begin{aligned} f_1'(x) &= \left(\sqrt{1+x^2} \right)' \frac{1}{\sqrt{1+x^2}} \\ &= \frac{2x \cdot \frac{1}{2} \left(\frac{1}{\sqrt{1+x^2}} \right)}{\sqrt{1+x^2}} \\ &= \frac{x}{1+x^2} \end{aligned}$$

b)

$$\begin{aligned} \left(\sqrt{1+\frac{1}{x}} \right)' &= -\frac{1}{x^2} \cdot \frac{1}{2} \cdot \frac{1}{\sqrt{1+\frac{1}{x}}} \\ f_2'(x) &= \left(\sqrt{1+\frac{1}{x}} - \frac{1}{2x\sqrt{1+\frac{1}{x}}} \right) \cdot \exp\left(x\sqrt{1+\frac{1}{x}}\right) \end{aligned}$$

c)

$$\begin{aligned} f_3'(x) &= 2x\sqrt{x+1} + (x^2+2) 2x \cdot \frac{1}{2} \cdot \frac{1}{\sqrt{x+1}} \\ &= 2x\sqrt{x+1} + \frac{(x^2+2)x}{\sqrt{x+1}} \end{aligned}$$

d)

$$f_4'(x) = -\sin(x) \cdot \cos^2(x)$$

e)

$$\begin{aligned} f_5'(x) &= \frac{\frac{\exp(\sin(x))}{x} + \ln(x) \cdot \cos(x) \cdot \exp(\sin(x))}{\exp(\sin(x))^2} \\ &= \frac{\frac{1}{x} + \ln(x) \cos(x)}{\exp(\sin(x))} \end{aligned}$$